

Andrew Burkard

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Work Experience

HackerPulse

HEAD OF AI

San Francisco, CA

Apr 2025 - Present

- **AI Agent Platform** – Designed and built the AI service that powers the product. Engineering leaders chat with an AI agent connected to live data from GitHub, Jira, Slack, Notion, etc. to get answers about their engineering org.
- **Knowledge Graph** – Built a self-reinforcing knowledge graph across all synced integrations — agents discover implicit relationships, each new connection helps surface the next, and the user-facing agent queries it via tool calls.

Yuma AI (YC W23)

FOUNDING AI ENGINEER

San Francisco, CA

Apr 2023 - Dec 2024

- **Cost Optimization** – Fine-tuned Mistral 7B to replace GPT-4 for customer support ticket classification, which runs on every incoming ticket, cutting per-call inference costs by ~100x. Created an LLM-as-judge evaluation system to safely validate prompt changes and migrate between model generations as new ones launched.
- **Smart Classification** – Designed an active learning loop for ticket routing: embed tickets → SVM classifies known types → anomaly detector catches out-of-distribution tickets → LLM labels the unknowns → retrain.
- **Hybrid RAG** – Implemented the retrieval pipeline combining embedding search with BM25-style keyword matching.

Triplebyte

MACHINE LEARNING ENGINEER

San Francisco, CA

Apr 2019 - Mar 2023

- Joined as the first ML hire; promoted to Tech Lead of Data & Infrastructure.
- **Search** – Migrated candidate search from a Postgres read replica (20 seconds per search on a bad day) to Elasticsearch (60-300ms). Also enabled hybrid search: BM25, metadata filters, and vector similarity in one index.
- **Responsiveness** – Trained a simple, but unreasonably effective logistic regression to predict which candidates would respond to recruiter outreach (three features: past response rate, email opens, recent logins) that doubled the response rate from 20% to 40%. Recruiters loved it; our sales team less so, since we charged per message.
- **Matchmaking** – Crafted a recommendation engine for a two-sided recruiting marketplace using neural collaborative filtering: content-based features handle cold start, then collaborative filtering takes over as interaction data accumulates.

B23

DATA SCIENTIST

McLean, VA

Oct 2016 - Apr 2018

- **Location Data Pipeline** – Built an ETL pipeline aggregating and resolving millions of points-of-interest from disparate sources for predictive foot-traffic analysis. Sold to hedge funds.
- **Cloud Migration** – Automated the transfer of hundreds of terabytes of imagery data between the AWS cloud and Snowball Edge devices running GIS software on-device for the US Navy, replacing a tedious process that involved burning thousands of DVDs.

Agilex / Accenture Federal Services

SOFTWARE ENGINEER

Chantilly, VA

Jun 2013 - Oct 2016

- **Pluto** – Developed a semantic search application for matching intelligence reporting with collection requirements via latent semantic indexing (LSI). Optimized the search engine around typical user behavior to increase the average query speed by 10x.

Education

Georgetown

M.S. DATA SCIENCE

Washington, D.C.

Sep 2016 - Aug 2018

Virginia Tech

B.S. COMPUTER SCIENCE AND MATHEMATICS, *summa cum laude*, PHI BETA KAPPA

Blacksburg, VA

Aug 2009 - May 2013

Projects

- **March Madness** – Annual Kaggle competitor since 2019. Developed a fully Bayesian hierarchical model in NumPyro/JAX (think KenPom, but probabilistic), adjusting for travel distance, altitude, and referee tendencies. Best finish: gold medal in 2021 (4th/451), the one year they didn't award a cash prize, so all I got was an (admittedly pretty cool) water bottle.
- **Language Models** – Trained ~100M parameter transformers in 2018 using data from ProFootballTalk comments, then Reddit, before "GPT-1" was even a thing. Deployed as Twitter bots on t2.small CPU instances, generating surprisingly coherent replies in under 2 seconds. Elon Musk even followed one briefly — then he bought the platform, promising to ban all the bots, so I take some blame there.

Skills

Languages

Python | Ruby | TypeScript | SQL

AI/ML

PyTorch | JAX | NumPyro | OpenAI Agents SDK | LangChain | MCP | LoRA | Hugging Face | scikit-learn | XGBoost | pandas

Infrastructure

PostgreSQL | FastAPI | Elasticsearch | Redis | AWS

Interests

Soccer | Prediction markets | Sports analytics | Jeopardy | Britpop | Seinfeld